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# Deterministic Evacuation Mapping: An Automated QGIS Plugin for Building-Level Analysis

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**Software**

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## Summery

**Build_Short_Evac_Time** is an open-source QGIS plugin that supports deterministic **building-level evacuation planning** for flood and inundation hazards. It automatically identifies buildings located within an inundation zone and computes **shortest evacuation routes** from those at-risk buildings to nearest safe shelters using open street map road network data (OpenStreetMap contributors, 2024). The plugin aims to assist GIS users and disaster risk planners in generating rapid evacuation path outputs with minimal input requirements

## Statement of need

Evacuation planning in flood and inundation hazard-prone areas (e.g., tsunami, storm surge and coastal flooding) requires tools that can efficiently compute and visualize evacuation routes from individual buildings toward safe locations. Although existing GIS workflows can perform network-based analysis and QGIS (QGIS Development Team, 2024) provides general spatial and network analysis capabilities, implementing a complete workflow for building-level evacuation routing typically requires multiple manual steps or custom scripting. There is a lack of easy-to-use QGIS plugins tailored specifically for at-risk building identification and shortest time evacuation assessment based on walk-time in inundation scenarios.

**Build_Short_Evac_Time** a QGIS plugin addresses this gap by integrating inundation-based buildings at risk selection with time-based network routing in a single interface. This plugin provides an open-source implementation of that framework within QGIS, enabling reproducible and operational evacuation planning. The plugin computes evacuation time rather than only distance and assigns each building to the nearest shelter that can be reached in the shortest time.

This tool supports disaster risk planners, local goverment, researchers and GIS practitioners in generating reproducible and terrain-aware evacuation analyses within an open-source GIS environment.

## Methodology

The plugin follows an automated, deterministic workflow that identifies at-risk buildings by overlaying building footprints with the inundation extent. Safe shelters are then identified, and the OpenStreetMap pedestrian network is used to calculate the shortest evacuation paths from buildings to shelters. Where a DEM is provided, terrain slope is incorporated using Tobler's Hiking Function (Tobler, 1993) to estimate walking time. The resulting evacuation time is

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stored as an attribute and buildings can be classified according to evacuation time thresholds for further analysis and visualization. The resulting classified buildings and evacuation routes are generated as QGIS output layers. The theoretical framework of the tsunami walk-time modelling approach is described in Khairnar, G. et al.(Khairnar et al., 2026).

## Functionality

The plugin operates within the QGIS environment and provides an integrated workflow for building-level evacuation time analysis. Its main functionality includes:

- **Input layers**:
  - Building footprints (Polygon vector layer in .shp or .geojson format)
  - Inundation or hazard extent polygon (Polygon vector layer in .shp or .geojson format)
  - Shelter locations (Point vector layer in .shp or .geojson format)
  - Optional Digital Elevation Model (Raster layer in preferably in .tiff format) for slope-based walking speed adjustment
- **At-risk building identification**: Buildings intersecting the inundation polygon are automatically detected and selected for evacuation analysis.
- **Time-based network routing**: For each at-risk building, the plugin computes evacuation routes to available shelters using the OSM road network. Evacuation time is calculated based on walking speed assumptions.
- **Minimum-time shelter allocation**: Each building is assigned to the shelter that can be reached in the shortest evacuation time rather than shortest distance.
- **Optional terrain-adjusted walking speed**: When a Digital Elevation Model (DEM) is provided, slope values along the evacuation route are used to adjust walking speed, allowing terrain-dependent evacuation time estimation.
- **Output generation**: After successful execution, 2 output layers will generate, first is buildings which are at risk and second is evacuation routes. Each building is assigned to the nearest evacuation shelter and that shelter's ID and evacuation time is added in that building's attribute. Buildings are classified according to evacuation time categories for further spatial analysis and visualization.

## Usage and Outputs

Figure 1 illustrates the plugin interface and Figure 2 represents its outputs. The user selects the building footprint as polygon layer, evacuation shelter as point layer and inundation extent layers as polygon layer, with an optional digital elevation model (DEM) in tiff format and clicks Run to execute the analysis. The resulting map displays buildings classified by estimated evacuation time (in minutes), along with the shortest evacuation routes and their directions toward assigned shelters shown in Figure 2a. The building attribute table provides the corresponding shelter assignment (id_assigned), estimated walking time (Time) and evacuation time class (Class) for each building mentioned in Figure 2b.

- **Input Parameters**

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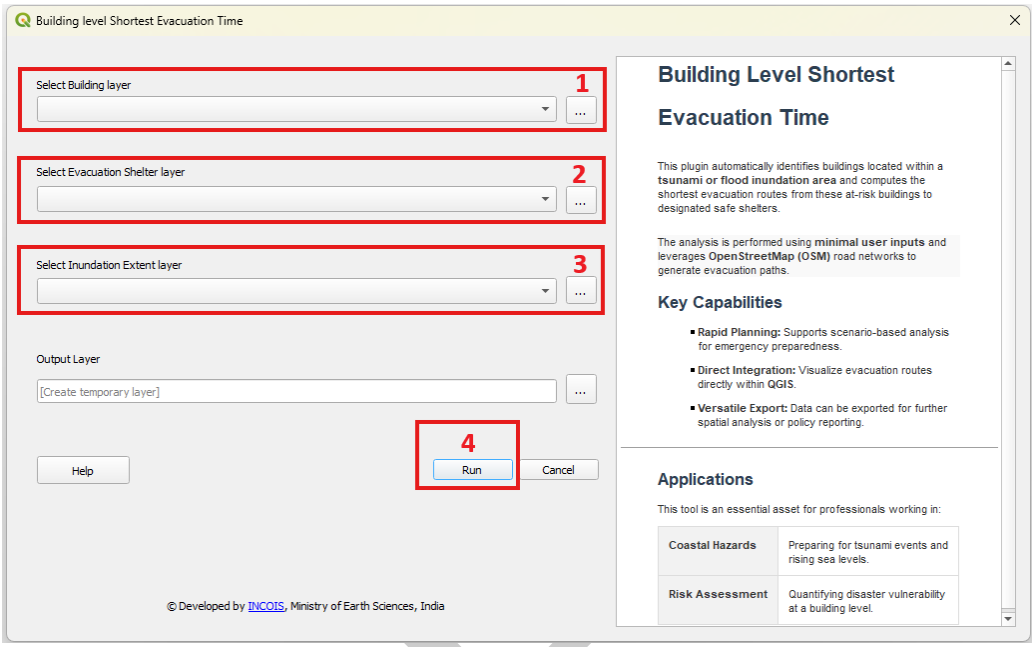



**Figure 1:** Input Parameters

78 ▪ **Output Layers**

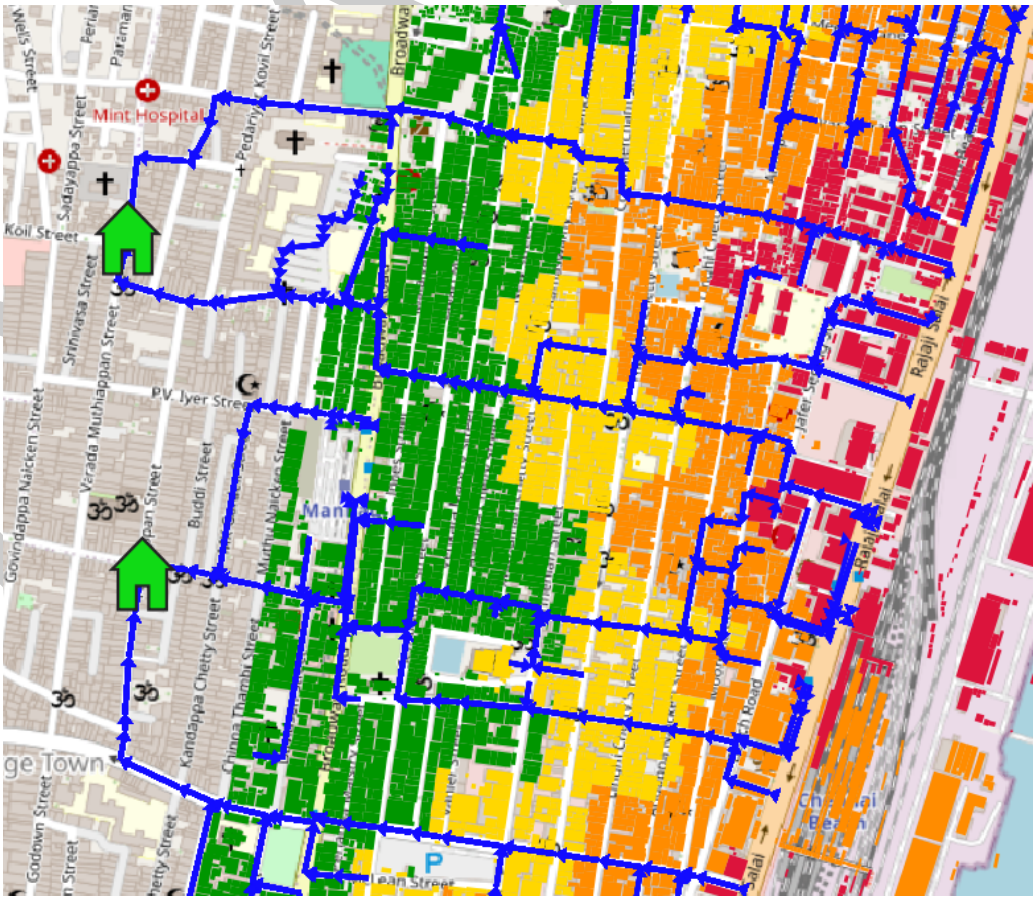
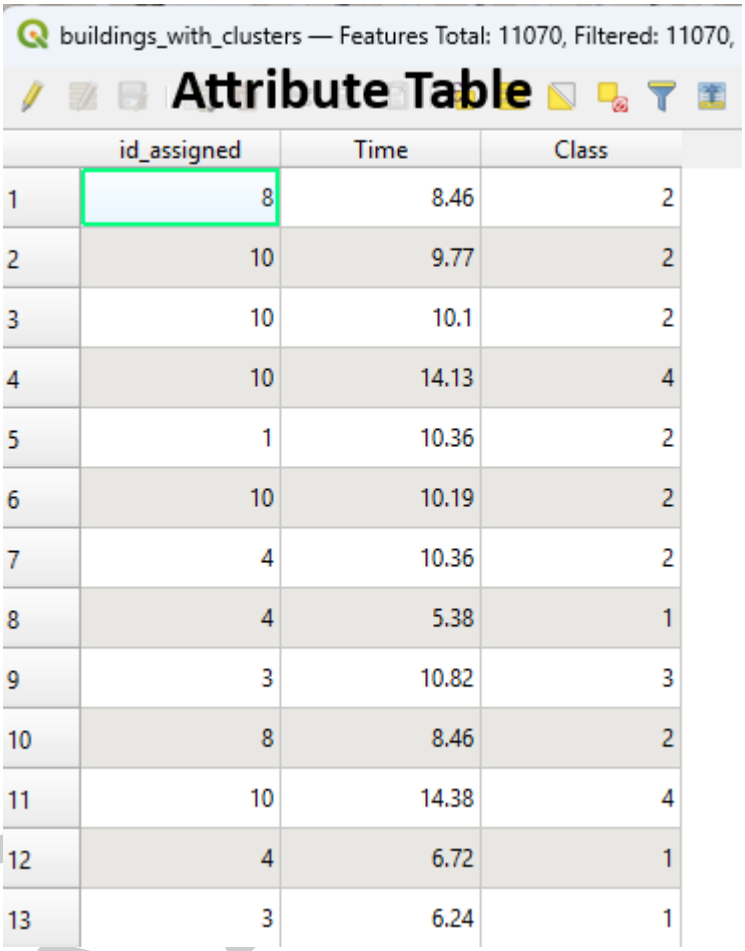



**Figure 2:** Output Layers

- **Attributes**




| | id_assigned | Time | Class |
|---|---|---|---|
| 1 | 8 | 8.46 | 2 |
| 2 | 10 | 9.77 | 2 |
| 3 | 10 | 10.1 | 2 |
| 4 | 10 | 14.13 | 4 |
| 5 | 1 | 10.36 | 2 |
| 6 | 10 | 10.19 | 2 |
| 7 | 4 | 10.36 | 2 |
| 8 | 4 | 5.38 | 1 |
| 9 | 3 | 10.82 | 3 |
| 10 | 8 | 8.46 | 2 |
| 11 | 10 | 14.38 | 4 |
| 12 | 4 | 6.72 | 1 |
| 13 | 3 | 6.24 | 1 |

**Figure 3:** Attribute Table

These outputs can be visualized and further analyzed within QGIS to support building-level evacuation planning. Users can also export the generated layers as vector files, such as shapefiles, for permanent storage, integration with other GIS datasets and further spatial analysis using external GIS software. The building-level attributes, including assigned shelter IDs, estimated walking times, and evacuation time classes, enable users to identify areas with longer evacuation times, examine shelter service areas and support the preparation of evacuation plans and disaster management reports.

## Availability

The source code for Building-level Shortest Evacuation Time is openly available at: https://github.com/gauravbkhairnar/Build_Short_Evac_Time_QGIS_Plugin

The plugin is distributed under the MIT License. It is designed to operate within the QGIS environment and can be installed through the QGIS Plugin Manager or manually from the repository. The project documentation, installation instructions and example usage are provided in the repository README file.

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